Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Drawing a Parallel Line Through a Point

Two lines are parallel when they run side by side and never meet, no matter how far they are extended. You are going to draw a line that passes through a given point P and stays parallel to a given line AB. You will need only a set square and a ruler from your mathematical set.

Example: Construct a line through the point P so that it is parallel to AB.

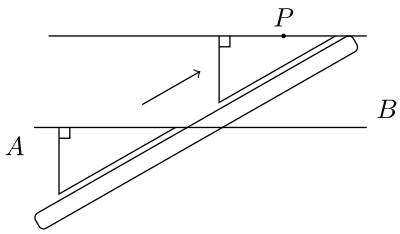
Step 1. Place the set square so that one of its edges lies exactly along the line AB (fig-01).

Step 2. Place the ruler along one of the other edges of the set square. If you are right-handed, use the left-hand edge (fig-02). You may rest the ruler against the set square's sloping edge instead of its upright one (fig-03) — either works, and the ruler's slant does not affect the construction. What matters is that one edge of the set square lies along AB and the ruler rests against a different edge of the set square.

Step 3. Hold the ruler firmly so it does not move. Slide the set square along the ruler, towards P. Sliding along a held ruler moves the set square without turning it, so its edge keeps the same direction (fig-04 and fig-05). Stop when the edge that was lying on AB reaches P.

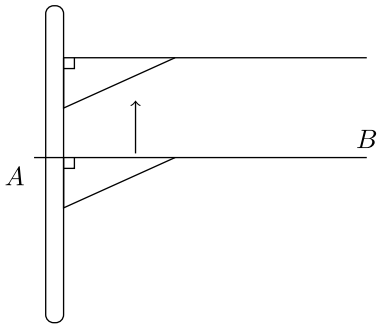
Step 4. Draw a line along that edge of the set square, through P. Because the set square was slid, not turned, this edge keeps the same direction as AB, so the line you draw through P is parallel to AB.

Visual



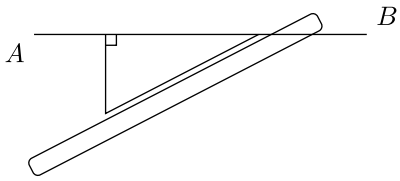
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



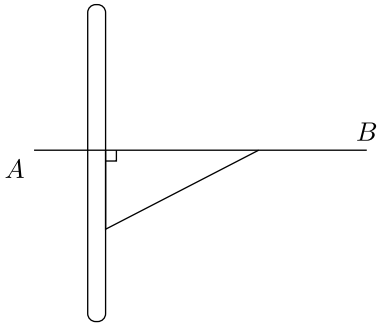
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



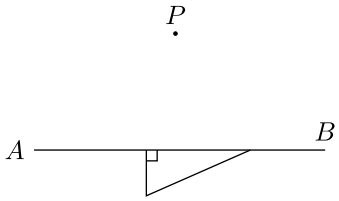
The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Practice

1. Draw a horizontal line AB about 8 cm long with a point P marked about 3 cm above it. Construct a line through P that is parallel to AB, using only your set square and ruler.
2. You have placed the set square with one edge along AB and the ruler against another of its edges. Before drawing, you must do two things in order. What are they?

Section 2: Constructing Perpendicular Lines

Two lines are said to be **perpendicular** to each other when they meet at a right angle — that is, at 90 °. In fig-06 you can see line AB crossed by line XY; the small square drawn at the crossing marks the right angle, so XY is perpendicular to AB.

To construct a perpendicular line yourself, you need only your ruler and set square. Here is the way to do it.

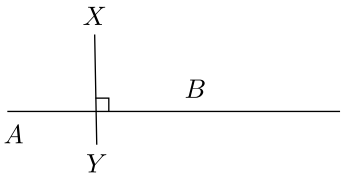
Step 1 — Draw the line and mark the point. Draw a line AB and mark a point Q somewhere on it, as in fig-07. This point Q is where your new perpendicular line will cross AB.

Step 2 — Place the set square. Put the edge of the set square on line AB. Now place your ruler so that its edge meets the hypotenuse (the long sloping edge) of the set square, as in fig-08. Notice how the set square stands upright on AB, with its right angle sitting on the line.

Step 3 — Slide and draw. Pressing firmly on the ruler so it cannot move, slide the set square down along the hypotenuse until the third edge of the set square touches point Q. Now draw a line XY along that third edge, letting X lie above AB and Y below it, as in fig-09.

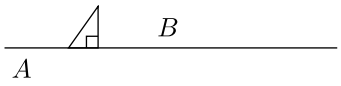
Because you slid the set square without turning it, its right angle is preserved, so the line you draw crosses AB at exactly $90\,^\circ$. The two lines AB and XY are perpendicular.

Visual



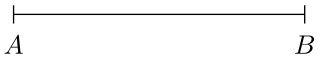
Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Visual



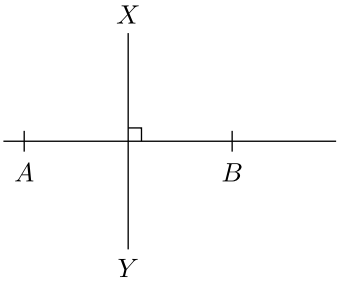
Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Example — construct a line perpendicular to PQ at the point R on it.

1. Draw line PQ and mark point R on it.
2. Lay the edge of the set square along PQ, then place the ruler against the hypotenuse of the set square and hold it firm.
3. Pressing down on the ruler, slide the set square along the hypotenuse until its third edge touches R.
4. Draw line XY along that third edge so that X is above PQ and Y is below it.

Check your result: the small right-angle mark at the crossing of PQ and XY shows the two lines meet at $90\,^\circ$, so they are perpendicular.

Practice

1. Draw a line AB and mark a point Q on it. Using your ruler and set square, construct a line through Q that is perpendicular to AB. What do you look for to confirm that your line is perpendicular?
2. Two lines XY and UV meet at a point and the angle between them is $90\,^{\circ}$. In your own words, what do you call these two lines?

Section 3: Bisecting a Line Segment

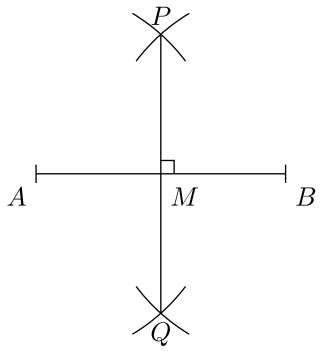
To bisect a line segment means to divide it into two equal halves. The point where the two halves meet is called the midpoint, and it lies exactly in the middle of the segment. Here is how you bisect a given line segment with a pair of compasses and a ruler.

Start with your segment, say AB (fig-10). You will need a pair of compasses, a ruler and a sharp pencil.

1. Open your compasses so that the radius is about three-quarters of the length of AB. The exact opening is not critical, but it must be more than half of AB, otherwise the arcs below will not cross.
2. Place the sharp point of the compass on A. Draw two arcs, one above the line and the other below the line, near the middle of AB.
3. Keeping the compasses at the same radius, move the sharp point to B. Draw two more arcs so that they cut the first pair of arcs. The upper pair of arcs crosses at a point P, and the lower pair crosses at a point Q (fig-11).
4. With your ruler, join P to Q. The line PQ crosses AB at a point M (fig-12).

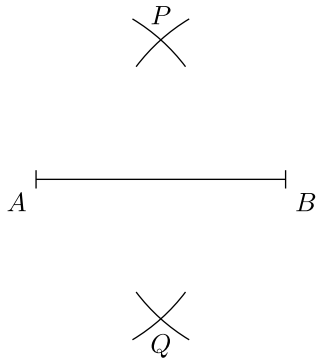
The point M is the midpoint of AB. That means the two halves are equal: AM = MB. So you have cut the segment into two equal halves, and PQ meets AB at a right angle, 90°.

Visual



The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The line segment AB, a stroke at each end, A on the left and B on the right.

Worked example — bisecting segment CD

Suppose you are asked to bisect the segment CD. Follow the steps in order.

1. Draw the segment CD, with C on the left and D on the right.
2. Open the compasses to about three-quarters of the length of CD — more than half of it.
3. Put the sharp point on C. Draw an arc above the line and an arc below the line, near the middle.
4. Keep the same radius. Move the sharp point to D. Draw an arc above the line and an arc below the line, so that they cut the first arcs. Mark where the upper arcs cross as P and where the lower arcs cross as Q.
5. Join P to Q with a ruler. Where PQ meets CD, mark the point M.

M is the midpoint of CD, so CM = MD, and the two halves are equal. PQ meets CD at a right angle.

Practice

1. Bisect the line segment AB shown in fig-10. What does the word "bisect" mean, and what must be true of the two halves after you finish?
2. In the construction in fig-12, the line PQ crosses AB at M. What does the point M divide AB into, and at what angle does PQ meet AB?
3. When bisecting a segment, your compasses must be open to more than half the length of the segment. Why is that important?

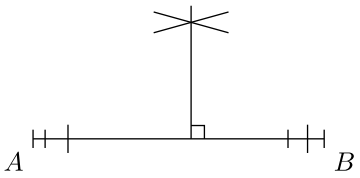
Section 4: Constructing a 90° Angle

A right angle — an angle of exactly 90° — is one you can build for yourself with just a ruler and a pair of compasses. The trick is that a perpendicular line and a right angle are the same thing: if a line rises straight up from a point on another line so that the two are perpendicular, the angle where they meet is 90°. So to construct a 90° angle you construct a perpendicular.

Look at fig-13. A straight line AB has been drawn, and a point C marked on it where the right angle is wanted. A perpendicular rises from C, and where it meets the line there is the small square mark that tells you the angle is a right angle. The two short strokes crossing each other near the top of the perpendicular are construction marks from the compass — they show the steps used to build it. You will make those same marks yourself.

What matters is the order: first place the point C on the line, then sweep one arc from C, then two wider arcs from the points that arc makes, and finally join C to where those two arcs cross.

Visual



A line with a stroke at each end, A on the left and B on the right, and four short strokes standing on the line, a pair towards A and a pair towards B. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Construct a 90° angle at a point C on the line AB, and follow along with fig-13.

1. Draw a straight line AB and mark a point C on it where the 90° angle is to be constructed.
2. Put the point of the compasses at C and draw an arc that cuts the line at two points, D and E — one on each side of C. These two points are now marked on the line, at equal distances from C.
3. Open the compasses WIDER than DC (the width used in step 2 will not work, because arcs drawn at that width from D and E meet only at C itself). With the compasses at D at this new, wider setting, draw an arc above the line.
4. Keep the compasses at the same wider setting from step 3, place them at E, and draw another arc above the line. This second arc cuts the first one at a point F.
5. Join C to F with a ruler.
6. CF is perpendicular to AB, so the angle formed between CF and the line at C is a right angle: $\angle FCB = 90°$.

The one step to be careful with is the third: the second setting of the compasses must be wider than the distance DC. That is exactly why the two arcs from D and E cross each other at F instead of meeting at C — the wider setting pushes their crossing point up above the line, and joining that point back to C gives you a line standing straight up at 90°.

Practice

1. Construct a 90° angle at a point C on a line AB, and write down, in the correct order, the two widths you set the compasses to.
2. In the construction of a 90° angle, why must the compasses be opened wider for the second set of arcs than for the first?

Section 5: Constructing an Angle of 60°

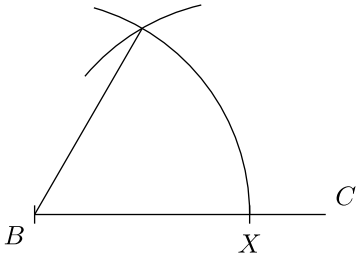
You can construct a 60° angle using only a ruler and a pair of compasses, by drawing arcs of equal radius. Here is the idea: if you swing two arcs of the same radius so that one cuts the other, the crossing point lies exactly on the ray that makes 60° with your original line.

Start with a line segment BC, with B on the left. Place the point of your compasses on B, open them to any convenient radius, and swing an arc that cuts BC at a point X (see fig-14). Keep this same radius — it stays fixed for the whole construction, and that equal radius is what makes the angle work out to 60°.

Now place the point of your compasses on X and, with the same radius as before, swing a second arc that cuts the first arc. Call the point where the two arcs cross A.

Finally, draw the straight line from B to A. The angle ABC is a 60° angle.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Let us build a 60° angle step by step.

Step 1 — Draw a line segment and mark B. Label the left end B and the right end C.

Step 2 — With your compasses' point on B, choose any convenient radius and swing an arc that cuts BC. Mark where it cuts BC as X.

Step 3 — Without changing the radius, move the compasses' point to X and swing a second arc that cuts the first arc. Label the crossing point A.

Step 4 — Join B to A with a straight line. The angle ABC (between BA and BC) is 60°.

Check your work in fig-14: the long arc from B through X, the shorter arc from X cutting it, and the straight line from B up to A. Your three points B, X and A are all the same distance apart because the radius never changed — and a triangle whose three sides are equal always has a 60° angle at each corner.

Practice

1. Write down, in the correct order, the steps you follow to construct a 60° angle at B on a line segment BC.
2. In constructing a 60° angle, you swing a second arc from X using the same radius as the first arc. Why must the two radii be equal?

Check your work

Attempt every question before you read this page.

Drawing a Parallel Line Through a Point

1. The line through P runs horizontally (parallel to AB), never meeting AB however far extended. Check it by eye: it is the same distance from AB all the way along.

   Place one edge of the set square exactly along AB. Rest the ruler firmly against another edge of the set square. Hold the ruler still and slide the set square along it until the edge that was on AB reaches P. Draw a line along that edge through P. Since the slide moved the set square without turning it, the drawn line keeps the direction of AB.
2. First hold the ruler firmly so it cannot move, then slide the set square along the ruler until the edge that was on AB reaches P. Only then draw the line along that edge through P.

Constructing Perpendicular Lines

1. The constructed line meets AB at a right angle ($90\,^\circ$), shown by the small right-angle mark at the crossing.

   Lay the edge of the set square along AB, hold the ruler against the hypotenuse, and slide the set square until its third edge touches Q. Draw the line along that third edge. The two lines meet at $90\,^\circ$, so they are perpendicular.
2. Perpendicular — two lines that meet at a right angle ($90\,^\circ$) are perpendicular to each other.

Bisecting a Line Segment

1. To bisect means to divide into two equal halves. After bisecting, the two halves are equal, so AM = MB, where M is the midpoint.

   Open the compasses to about three-quarters of the length of AB (more than half). Draw two arcs from A, one above and one below the line. Keeping the same radius, draw two arcs from B so that they cut the first arcs at P (above) and Q (below). Join P to Q with a ruler. The crossing point M on AB is the midpoint, so AM = MB.
2. M is the midpoint of AB, so it divides AB into two equal halves AM and MB. PQ meets AB at a right angle, 90°.
3. If the compasses were open to less than half the length, the arcs drawn from the two ends would not cross each other above and below the line, so you would not get the crossing points P and Q to join up.

Constructing a 90° Angle

1. The two widths are: first, the width used to draw the arc from C that cuts the line at D and E; second, a WIDER setting (wider than DC) used for the arcs from D and E that cross at F. The final angle at C is 90°.

   After drawing line AB and marking C, set the compasses at C and draw an arc cutting the line at D and E (width 1). Then open the compasses wider than DC (width 2), put the point at D and draw an arc above the line, then keep width 2 and do the same from E so the arcs cross at F. Join C to F; CF is perpendicular to AB, so ∠FCB = 90°.
2. Because the two wider arcs need to meet at a point F above the line. If the compasses stayed at the first width (DC), the arcs drawn from D and E would meet only at C itself — right on the line — and that would not give a perpendicular rising from C.

Constructing an Angle of 60°

1. a. Swing an arc from B with any convenient radius to cut BC at X.
   b. With the same radius, swing an arc from X to cut the first arc at A.
   c. Join B to A. The angle ABC is 60°.
2. Because equal radii make the three points B, X and A all the same distance apart, forming an equilateral triangle whose angles are each 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-022e8f35	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____